

When Harm Scales: Measuring Critical Regimes in Financial Agent Societies

Anonymous submission

Abstract

AI agents are moving from isolated assistants to interconnected participants in markets, platforms, and organizations, where small harmful minorities can alter shared state and induce correlated behavior from benign agents. We study this failure mode as harmful-agent scaling. The central claim is that harmful-agent safety has a scaling dimension: the key quantity is not only whether an individual agent is harmful, but the critical population fraction at which society-level collapse emerges. We formalize this as a finite-size critical-regime measurement problem and introduce testable estimands rather than a claim of universal phase transition: the critical harmful-agent ratio $\alpha_c(N)$, transition width, mechanism-specific thresholds, structural threshold shifts, and defense-induced threshold shift. WolfBench is the instrument for measuring these quantities: a controlled financial agent-society environment that sweeps harmful-agent ratios, society sizes, manipulation mechanisms, and defenses under repeated seeds. This reframes agent safety from isolated-agent evaluation toward collective-risk measurement, where useful defenses should shift the estimated collapse threshold upward while preserving clean-market utility.

Introduction

AI agents are moving from isolated assistants to interconnected participants in workflows, markets, platforms, and organizations. In these settings, actions are coupled through shared state. One agent’s message can change what others believe; one agent’s trade can change the market observed by others; and a platform-level intervention can reshape the information and action space faced by many agents simultaneously. Harm is therefore not only a property of one request, one answer, or one trajectory. A small harmful minority can change the shared environment, causing benign agents to amplify the original signal and produce system-level damage.

This paper studies this collective failure mode through the lens of *harmful-agent scaling*. The basic question is simple: as the fraction α of harmful agents increases in a finite agent society, how does the probability of collapse change? This question adds a scaling dimension to harmful-agent safety.

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In many evaluations, harmfulness is measured at the level of a single agent: whether it follows an unsafe instruction, performs a harmful task, or violates a rule. Those evaluations are necessary, but they miss a different quantity: the critical population fraction at which individually small harmful actions become a society-level failure. Even if each individual action is small, a coordinated harmful minority can create correlated exposure, drive correlated benign behavior, and move the whole system into a risky regime.

Financial agent societies make this problem concrete. Harmful agents may manipulate social exposure, distort perceived demand, or occupy influential positions in a communication graph. Benign retail agents then convert public signals into trades. Market impact can feed back into the shared information state: price movement changes what agents believe, while spreading messages induce more correlated behavior. The question is therefore not only whether a harmful action can be detected, but whether the society has a measurable critical regime in which a small change in adversarial composition causes a large change in collapse probability.

The contribution is therefore not merely a new benchmark. We argue that harmful-agent safety should be measured as a finite-size critical-regime problem. WolfBench is the measurement instrument that makes this claim testable in a controlled financial agent society. It lets us sweep the harmful-agent ratio, repeat controlled episodes over random seeds, estimate collapse probability curves, and compare defenses by whether they shift the critical harmful-agent ratio upward. This reframes agent safety from isolated-agent evaluation toward collective-risk measurement.

Our contributions are:

- We identify a scaling dimension of harmful-agent safety: beyond individual harmfulness, the key society-level quantity is the critical harmful-agent fraction at which collapse emerges.
- We formulate harmful-agent scaling as a finite-size critical-regime measurement problem with estimands for collapse probability, critical ratio, transition width, mechanism heterogeneity, and structural threshold shift.
- We introduce WolfBench as a controlled measurement instrument that instantiates social, market, and intervention feedback loops so the scaling estimands can be measured reproducibly.

Figure 1 placeholder: main harmful-agent scaling result.

Panel A: harmful minority $\rightarrow$ exposure $\rightarrow$ benign amplification $\rightarrow$ market feedback $\rightarrow$ collapse.

Panel B: raw $P(C = 1 \mid N, \alpha)$ transition curves for multiple N .

Panel C: finite-size data collapse using $s_N(\alpha - \alpha_c(N))$.

Panel D: $\alpha_c(N)$ decreases with N with bootstrap confidence intervals.

Panel E: transition width W_N narrows with N .

Figure 1: Overview of harmful-agent scaling. The scientific object is a finite-size critical regime: harmful minorities inject correlated pressure, benign agents amplify it through social and market feedback, and collapse probability changes sharply near an estimated critical population fraction $\alpha_c(N)$. WolfBench is the instrument used to measure this quantity in a controlled agent society. The final version of this figure will combine the closed-loop schematic, transition curves, finite-size data collapse, critical ratio scaling, and transition-width scaling.

- We propose defense evaluation by threshold shift, $\Delta\alpha_c = \alpha_c^{\text{defense}} - \alpha_c^{\text{noguard}}$, together with clean-market and false-positive costs.

Problem Setup

We consider a finite-horizon agent society with N agents and horizon T . For a finite society, the benchmark instantiates $K \in \{0, \dots, N\}$ harmful agents and writes $\alpha = K/N$. When we refer to an α -grid, each requested ratio is implemented by the nearest feasible harmful-agent count. The remaining agents are benign retail agents and market participants. A mechanism $m \in \mathcal{M}$ specifies the harmful behavior family.

At each time t , the environment has a market state x_t , a social state g_t , and agent portfolios p_t . Agents observe public summaries of this state and choose messages or trades. Let μ_b denote the benign-agent policy class, let μ_h^m denote the harmful policy for mechanism m , and let π denote a defense policy. The transition is stochastic:

$$(x_{t+1}, g_{t+1}, p_{t+1}) \sim \mathcal{P}_m(\cdot \mid x_t, g_t, p_t, N, K, \mu_b, \mu_h^m, \pi),$$

with $\alpha = K/N$. The no-defense baseline is $\pi = \emptyset$.

Collapse is an episode-level event. Let $z_t \in \mathbb{R}_+^q$ collect nonnegative risk components such as price dislocation, liquidity stress, retail loss, social cascade size, and wealth transfer. WolfBench reports a continuous severity score

$$S_t = w^\top z_t,$$

but the binary collapse event is triggered by calibrated component thresholds:

$$C = \mathbf{1} \{ \exists t \leq T, \exists j \in \{1, \dots, q\} : z_{t,j} \geq \tau_{j} \}.$$

The scalar score is used for continuous severity reporting; the binary event is used for threshold estimation. The main object of measurement is therefore

$$p_m(N, \alpha; \pi) = \Pr(C = 1 \mid N, K, m, \mu_b, \mu_h^m, \pi).$$

The probability is taken over random initial conditions, social-graph realizations, harmful-agent placement, stochastic agent decisions, and exogenous market noise.

For a fixed mechanism and defense policy, the critical harmful-agent ratio is the midpoint of a fitted collapse-response curve:

$$\alpha_c^m(N; \pi) = \inf \{ \alpha : p_m(N, \alpha; \pi) \geq 1/2 \}.$$

Because empirical rates can be non-monotone under finite sampling noise, we treat α_c as a response-curve estimand rather than a pointwise property of raw rates. In experiments, raw collapse rates are summarized by a monotone or logistic response fit, and uncertainty is estimated from repeated seeds. The transition width is the distance between the α values where the fitted curve reaches 0.1 and 0.9. A narrow width means that collapse probability changes sharply near α_c .

A defense is evaluated as an intervention on the same system, not as a static classifier. The primary defense estimand is

$$\Delta\alpha_c^m(N) = \alpha_c^m(N; \pi) - \alpha_c^m(N; \emptyset).$$

A useful defense has positive threshold shift and low clean-market cost. This matters because an intervention that blocks everything may reduce collapse but also destroy normal market behavior.

Harmful-Agent Scaling: Hypotheses and Estimands

This section provides a stylized amplification model and defines the estimands tested by WolfBench. It does not prove a universal phase transition. Instead, it separates four objects: a mechanism-level motivation, a response-curve estimator, finite-size summary models, and local comparative statics.

Amplification model. Let $E_t \geq 0$ denote harmful exposure in the society: harmful messages, displayed pressure,

fake volume, or other mechanism-specific signals that can influence benign agents. Let $D_t \geq 0$ denote market dislocation for the target asset, and write $y_t = (E_t, D_t)^\top$. Around the clean-market regime, we use the linearized expectation model

$$\mathbb{E}[y_{t+1} | y_t] = M_m(\alpha, \theta)y_t + u_m(\alpha, \theta),$$

where

$$M_m(\alpha, \theta) = \begin{bmatrix} r_m(\alpha, \theta) & \phi_m \\ \kappa_m \beta_s / L_m & 1 - \lambda_m \end{bmatrix}.$$

Here θ collects graph reach, harmful-agent placement, and retail susceptibility; r_m is exposure reproduction; ϕ_m is social-market feedback; L_m is effective liquidity; λ_m is market damping; and $\kappa_m \beta_s / L_m$ converts exposure into market dislocation. We assume $r_m(\alpha, \theta)$ and the input $u_m(\alpha, \theta)$ are nondecreasing in α , and increase with harmful-agent centrality, graph reach, and retail susceptibility.

Iterating the linearized model gives

$$\mathbb{E}[y_t] = M_m^t y_0 + \sum_{\ell=0}^{t-1} M_m^\ell u_m.$$

When the spectral radius of M_m is well below one, harmful exposure is damped. As it approaches the unit scale, the same harmful input can be amplified over the horizon through the summation term. This mechanism motivates a critical regime in finite experiments, but it is not by itself a proof that all simulated mechanisms undergo a sharp transition.

Response-curve estimator. For each finite N , mechanism m , and defense policy π , we summarize measured collapse rates with a logistic response curve:

$$p_m(N, \alpha; \pi) = \sigma(s_N^m(\pi) (\alpha - \alpha_c^m(N; \pi))),$$

where $\sigma(u) = (1 + \exp(-u))^{-1}$, $\alpha_c^m(N; \pi)$ is the critical ratio, and $s_N^m(\pi) > 0$ is local steepness. This logistic curve is only a response-curve estimator for summarizing finite-grid measurements; it is not a claim that the simulator's true response is exactly logistic. It gives two interpretable summaries:

$$\alpha_c^m(N; \pi) = \inf\{\alpha : p_m(N, \alpha; \pi) \geq 1/2\},$$

$$W_N^m(\pi) = \alpha_{0.9} - \alpha_{0.1} = \frac{2 \log 9}{s_N^m(\pi)}.$$

The benchmark is designed to test whether α_c is stable under repeated seeds and whether the transition width narrows as the society size grows.

Finite-size summaries. We summarize finite-size trends with the parametric forms

$$\alpha_c^m(N) = \alpha_\infty^m + a_m N^{-\nu_m}, \quad W_N^m = b_m N^{-\gamma_m},$$

with $b_m, \nu_m, \gamma_m > 0$ and $a_m \in \mathbb{R}$. These are summary models, not universal laws. For social-amplification mechanisms, we expect $a_m > 0$, meaning that the critical ratio decreases toward α_∞^m as N grows. Other mechanisms may have weaker or different finite-size trends, which is why WolfBench reports mechanism-specific thresholds rather than forcing one global exponent.

Comparative statics. Let $A_m(N, \alpha, \theta; \pi)$ denote a latent expected collapse pressure over the horizon, and let τ_m be a calibrated threshold. We connect this latent margin to collapse probability by assuming

$$p_m(N, \alpha; \pi) = F_m(A_m(N, \alpha, \theta; \pi) - \tau_m),$$

where F_m is monotone and $F_m(0) = 1/2$. Under this link, the critical ratio satisfies

$$A_m(N, \alpha_c, \theta; \pi) = \tau_m.$$

Assume A_m is continuously differentiable, the critical solution is locally unique, and $\partial A_m / \partial \alpha > 0$ at the critical point. The implicit function theorem gives

$$\frac{\partial \alpha_c}{\partial \theta_j} = - \frac{\partial A_m / \partial \theta_j}{\partial A_m / \partial \alpha}.$$

Thus any parameter that increases latent collapse pressure lowers the critical ratio, while any parameter that dampens pressure raises it. The model motivates the following testable comparative statics:

$$\frac{\partial \alpha_c}{\partial L} > 0, \quad \frac{\partial \alpha_c}{\partial G} < 0, \quad \frac{\partial \alpha_c}{\partial \beta_s} < 0, \quad \frac{\partial \alpha_c}{\partial \phi} < 0,$$

where G is graph reach or centrality gain, β_s is social susceptibility, and ϕ is feedback strength.

Defense as threshold shift. Finally, a defense policy can be analyzed as a change in the latent pressure curve. Suppose $A_\pi(\alpha, \theta)$ and $A_\emptyset(\alpha, \theta)$ are nondecreasing in α , both define threshold crossings under the monotone link above, and

$$A_\pi(\alpha, \theta) \leq A_\emptyset(\alpha, \theta) \quad \text{for all tested } \alpha.$$

Then the defense critical ratio is no smaller than the no-defense critical ratio:

$$\alpha_c(N; \pi) \geq \alpha_c(N; \emptyset).$$

If the defended curve never crosses the collapse threshold on the tested grid, WolfBench reports a right-censored lower bound beyond the largest tested α . This motivates threshold shift as the primary defense metric. Average loss reduction is not enough: a useful defense should move the system away from the near-critical region while preserving clean-market utility.

Testable predictions. The preceding formulation yields five benchmark-level predictions. **P1** For fixed N and mechanism m , collapse rates should admit a stable response-curve midpoint $\alpha_c^m(N)$ and transition width. **P2** For social-amplification mechanisms, increasing society size should reduce the estimated critical ratio and narrow the transition width. **P3** Different mechanisms should have different $\alpha_c^m(N)$, because they enter through different amplification channels. **P4** Parameters that dampen latent pressure, such as liquidity, should increase α_c , while parameters that increase graph reach, centrality, susceptibility, or feedback should decrease it. **P5** A useful defense should produce positive $\Delta \alpha_c = \alpha_c^{\text{defense}} - \alpha_c^{\text{noguard}}$ with small clean-market and false-positive cost. The experiments below test these predictions as finite-size benchmark claims, not as proofs of a universal phase transition.

Prediction	Estimand	Expected direction	Primary test
P1 critical response	$\alpha_c^m(N), W_N^m$	stable midpoint and width	S1 alpha/N sweep
P2 finite-size scaling	$\partial\alpha_c/\partial N, W_N$	lower α_c , narrower W_N for social amplification	society-size sweep
P3 mechanism heterogeneity	$\alpha_c^m(N)$ across m	mechanism-specific thresholds	S1–S4 threshold audit
P4 structural statics	$\Delta\alpha_c$ under levers	signs follow L, G, β_s, ϕ	targeted threshold shifts
P5 defense objective	$\Delta\alpha_c$, utility cost	positive $\Delta\alpha_c$, low cost	defense critical-regime sweep

Table 1: Theory-to-experiment map. The paper is organized around estimands rather than a single benchmark score: each empirical section tests one prediction implied by the harmful-agent scaling formulation.

WolfBench as Measurement Instrument

WolfBench is a controlled instrument for measuring harmful-agent scaling. Its purpose is not to win benchmark coverage by adding another simulator, and it is not intended to be a high-fidelity exchange simulator or a real-market forecasting tool. The design goal is narrower and more fundamental for safety evaluation: hold the agent society, harmful mechanism, defense interface, and randomization protocol fixed, then measure how collapse probability changes as the harmful population fraction varies. This makes WolfBench an empirical testbed for the finite-size estimands in Table 1.

Agent Society

An episode contains N agents over a horizon $T = 30$ days. For each requested α , WolfBench instantiates the nearest feasible harmful count $K \approx \alpha N$; the remaining agents are benign retail agents, with a small number of market makers providing liquidity. Retail agents are heterogeneous: they place different weights on fundamentals, momentum, social exposure, volume, warnings, and noise. This heterogeneity matters because collective failure is not caused by a single irrational agent, but by correlated updates among many ordinary agents exposed to the same public signals.

The environment contains three assets with different liquidity and volatility profiles. Harmful mechanisms target one asset, while benign agents and market makers interact with the full market state. A static social graph connects agents, and messages propagate locally with decay and reshare effects. Market movement feeds back into the social layer: positive returns can increase the effective intensity of bullish messages, which in turn can induce additional buying. This loop is intentionally stylized. Its purpose is to expose the measurement problem created by coupled agents, not to reproduce microsecond exchange dynamics.

Manipulation Mechanisms

WolfBench includes one clean scenario and four harmful mechanisms (Table 2). The harmful mechanisms are case-grounded simulations: each is abstracted from public financial-market enforcement patterns or widely documented market-abuse cases. They are not attempts to reconstruct a particular historical event, and the benchmark omits operational details that would make the scenarios usable as playbooks. Instead, the cases provide realistic failure channels. S1 abstracts microcap pump-and-dump dynamics; S2 abstracts social-media influencer scalping; S3 abstracts spoofing and layering pressure; and S4 abstracts wash-trading and

fake-liquidity behavior. The mechanisms are chosen because they enter the amplification model through different channels: social exposure, graph centrality, displayed liquidity, and volume-as-signal feedback.

Defense Interface

A defense is a WolfGuard policy. On each day, it receives a public summary containing market snapshots, social signals, and recent returns. It returns a per-asset intervention from a fixed action set: no action, warning, cooldown, or block. The evaluator owns the intervention costs and false-positive rules, so a submitted policy cannot improve its score by re-defining costs. Standard submissions do not receive harmful identities, hidden policies, or private oracle pressure. An oracle policy may be reported as a non-eligible upper bound, but it is not part of the competitive track.

This interface is deliberately more restrictive than a full simulator API. The defender must operate through the same compressed public information channel that an actual platform-level intervention would plausibly use. This prevents the defense benchmark from collapsing into hidden-state classification.

Trajectory Dataset

WolfBench can also export a JSONL trajectory dataset for learning defense models. Each row corresponds to one (episode, day, asset) observation and contains exactly the public summary exposed to a WolfGuard policy. For training splits, the exporter can include oracle actions, future collapse indicators, future maximum severity, and component-level collapse labels. Held-out splits hide these labels by default. This design lets researchers train simulator-based defenses without giving evaluation-time access to private simulator state.

Metrics and Evaluation Protocol

WolfBench reports both continuous severity and binary collapse. The continuous score is useful for diagnosis, but the main scaling estimands use a binary event so that α_c and transition width are well defined.

Collapse and Critical-Ratio Estimation

The per-day CollapseScore is a weighted sum of component risks. Binary collapse is triggered when any calibrated component threshold is crossed (Table 3). This separation matters: the score describes how severe an episode became, while the event defines the response curve $P(C = 1 \mid N, \alpha)$.

WolfBench measurement environment

public closed loop; hidden labels separated

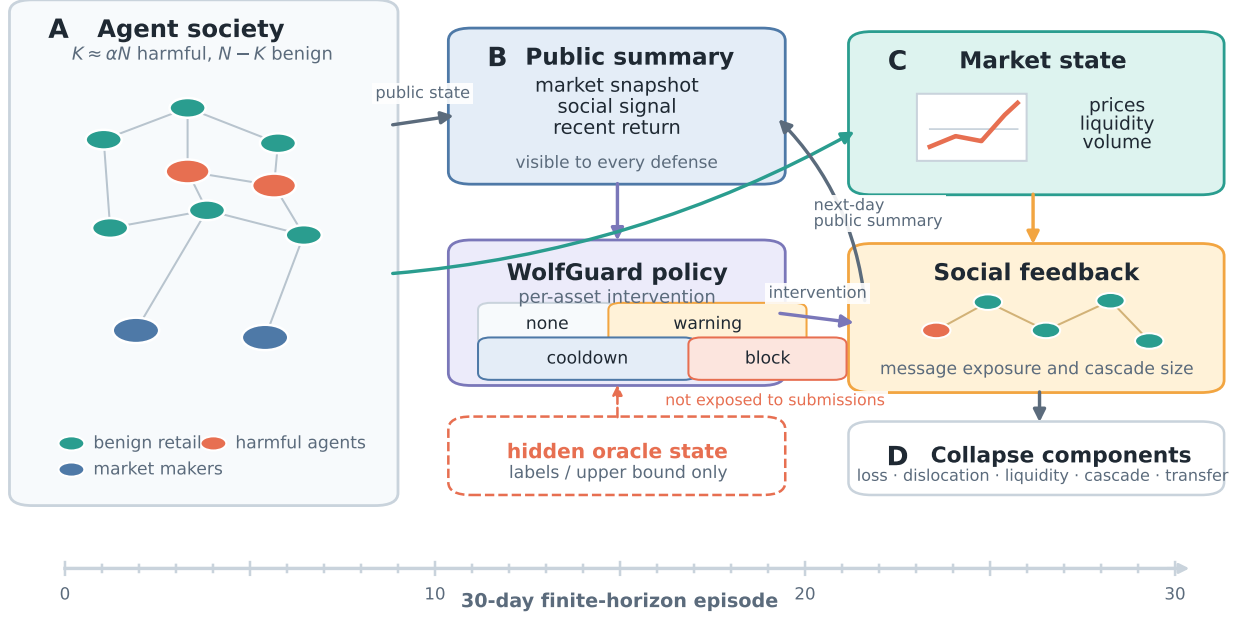


Figure 2: WolfBench measurement environment. Each episode runs a finite 30-day loop in which harmful agents inject mechanism-specific pressure, benign agents react to public social and market signals, the market updates prices and liquidity, and a defense policy can intervene using only public summaries. The benchmark therefore evaluates closed-loop collective risk rather than static classification.

We report Wilson intervals for raw collapse rates and bootstrap intervals for fitted α_c values.

For each N , mechanism m , and defense π , the benchmark sweeps an α -grid and fits a response curve. The primary statistics are the midpoint $\alpha_c^m(N; \pi)$ and the 10–90 transition width $W_N^m(\pi)$. These summaries are more informative than a single average harm score because they distinguish three regimes: clearly safe, near-critical, and collapsed. They also support threshold-shift comparisons across defenses and structural parameters.

Defense Scoring

Defense evaluation uses the same environment and seeds as the no-defense baseline. The main defense estimand is threshold shift,

$$\Delta\alpha_c = \alpha_c^{\text{defense}} - \alpha_c^{\text{noguard}}.$$

The benchmark also reports collapse reduction, retail-loss reduction, collapse delay, utility loss, false-positive rate, and intervention cost. The aggregate DefenseScore rewards harm reduction and positive threshold shift, but penalizes clean-market damage and excessive intervention. Harm reduction is gated by collapse reduction, delay, or positive threshold shift, so a policy cannot win by randomly suppressing market activity without improving safety. Because the current built-in defenses are diagnostic baselines rather than strong mitigations, the main paper treats defense leaderboards as

supplementary evidence; the core scientific claim is the scaling measurement itself.

Splits and Reproducibility

The benchmark defines public-dev, public-test, paper-final, hidden, and stress seed splits. Public-dev supports iteration and can include trajectory labels for training. Public-test uses held-out public seeds and hides labels. Paper-final uses a heavier seed profile for the figures and tables reported in the paper. Hidden and stress seeds are intended for official audits and robustness checks. Each experiment writes its configuration, per-episode rows, summary files, and run metadata, so reported figures can be traced back to concrete grids and seeds.

Experiments

The experiments are organized to test the predictions above. We separate estimand validation, mechanism breadth, parameter sensitivity, and defense evaluation so that each result answers a specific theoretical claim.

RQ1: Are Critical-Regime Estimands Stable?

This question tests P1 and P2 using the S1 society-size sweep. We estimate $P(C = 1 \mid N, \alpha)$ for $N \in \{100, 200, 500, 1000, 2000, 5000\}$, using a near-threshold α -grid and repeated seeds. We fit $\alpha_c(N)$, transition width, and

ID	Mechanism	Case grounding	Dominant channel	Role
S0	Clean market	Control setting without harmful agents	baseline drift and liquidity	calibration
S1	Social pump-and-dump	Public microcap and penny-stock pump-and-dump patterns	exposure, FOMO, price-social feedback	main scaling law
S2	Finfluencer scalping	Social-media promotion and follower scalping cases	high-centrality reach and copy trading	graph amplification
S3	Spoofing/layering	Public spoofing and layering enforcement cases	displayed depth, cancellations, liquidity stress	microstructure stress
S4	Wash trading	Crypto and thin-market wash-trading / fake-liquidity cases	fake volume and apparent liquidity	volume-signal stress

Table 2: WolfBench scenario cards. S1–S4 are case-grounded simulations abstracted from real financial-market abuse patterns, not reconstructions of specific historical events. The benchmark reports mechanism-specific thresholds rather than forcing one universal critical ratio across all failure families.

Component	Trigger	Interpretation
Retail loss	0.10	at least 10% loss of initial retail wealth
Price dislocation	0.30	target price deviates by 30% from fundamental value
Liquidity stress	3.00	spread/cancellation stress exceeds calibrated clean level
Social cascade	0.40	at least 40% of retail population exposed/adopted
Wealth transfer	0.05	harmful profit reaches 5% of initial retail wealth

Table 3: Collapse components. WolfBench reports a continuous weighted severity score, but the binary collapse event is triggered by calibrated component thresholds.

bootstrap confidence intervals. The key evidence is not a single collapse rate, but a family of response curves whose midpoint and width move systematically with N .

RQ2: Are Thresholds Mechanism-Specific?

This question tests P3 with cross-mechanism threshold estimation over S1–S4. We expect one estimator to apply across mechanisms, but not one universal critical ratio. S1 and S2 are expected to show cleaner social-amplification thresholds, while S3 and S4 are interpreted through their dominant liquidity and volume-signal channels.

RQ3: Do Structural Parameters Move Collapse Risk?

This question tests P4 using targeted threshold sweeps over structural levers: liquidity, graph degree, high-centrality placement, retail susceptibility, and social-market feedback. The preferred paper-facing estimand is the signed threshold shift $\Delta\alpha_c$ induced by each lever. Existing fixed-alpha sensitivity audits are useful diagnostics, but they are weaker evidence because they measure $\Delta P(C)$ at one operating point rather than movement of the response-curve midpoint. We therefore reserve the main structural figure for targeted $\Delta\alpha_c$ sweeps and place fixed-alpha audits in the supplement.

RQ4: Do Defenses Shift the Critical Ratio?

This question tests P5. We compare NoGuard, random controls, rule-based defenses, LLM defenses, and simulator-trained baselines on targeted critical-regime grids. The primary readout is $\alpha_c^{\text{defense}} - \alpha_c^{\text{noguard}}$, reported together with utility, false-positive, and intervention costs. Because the current built-in baselines are intentionally simple, the main text reports defense as a benchmark objective and moves the full leaderboard to the appendix unless a competitive defense produces a robust positive shift.

Results

We report results in the same order as the predictions, so the empirical section reads as a test of the theory-facing estimands rather than as unrelated benchmark tables.

Finite-Size Critical Regime

Figure 3 and Table 4 are the main evidence for P1 and P2. In the S1 sweep, the estimated critical harmful ratio falls from 0.0144 at $N = 100$ to 0.00937 at $N = 5000$. The 10–90 transition width also narrows from 0.0366 to 0.0183. A power-law summary gives $\alpha_c(N) \approx AN^{-\nu}$ with $\nu = 0.116$ and log-space $R^2 = 0.891$; the width fit gives $\gamma = 0.227$ and $R^2 = 0.837$. These fits should be read as finite-size summaries, not as universal exponents.

Mechanism-Specific Thresholds

Figure 4 tests P3. The important result is not that all mechanisms share one critical ratio. They do not. Instead, the same estimand reveals distinct critical regimes for distinct channels. S1 and S2 show cleaner threshold behavior under social amplification. S3 requires a wider grid at small N , and S4 exhibits broad transitions under the current composite collapse event. We therefore report S3 and S4 as mechanism-specific stress cases, not as failures of a universal law.

Structural Threshold Shifts

Figure 5 and Table 5 are reserved for the strongest test of P4: signed threshold shifts under structural levers. The theory predicts that increasing effective liquidity should raise α_c , while increasing graph reach, harmful centrality, retail susceptibility, or feedback should lower it. The final paper

Figure 3 placeholder: finite-size scaling law.

Panel A: $\alpha_c(N)$ with bootstrap confidence intervals and power-law summary fit.
 Panel B: transition width $W_N = \alpha_{0.9} - \alpha_{0.1}$ with width-scaling fit.
 Panel C: residual or stable-regime robustness fit for $N \geq 500$.
 Panel D: compact annotation of ν , γ , log-space R^2 , seed count, and alpha grid.

Figure 3: Finite-size scaling law for the S1 social-amplification mechanism. The main readout is not a single collapse rate, but the movement of the fitted response midpoint $\alpha_c(N)$ and the sharpening of transition width as the agent society grows.

N	α_c	95% CI	W_N	logistic slope
100	0.01439	[0.01278, 0.01610]	0.03662	119.99
200	0.01516	[0.01336, 0.01694]	0.04448	98.80
500	0.01377	[0.01232, 0.01533]	0.03613	121.62
1000	0.01236	[0.01115, 0.01355]	0.02348	187.16
2000	0.01126	[0.01008, 0.01240]	0.02081	211.13
5000	0.00937	[0.00817, 0.01049]	0.01826	240.62

Table 4: S1 society-size scaling summary. Each row is estimated from repeated 30-day episodes over an α -grid; confidence intervals are bootstrap intervals over seeds.

should report $\Delta\alpha_c$ with confidence intervals for each lever. Fixed-alpha sensitivity heatmaps are useful as diagnostics, but they should remain supplementary unless converted into threshold-shift estimates.

Defense Benchmarking as an Appendix Result

The defense track evaluates whether a policy moves the system away from the near-critical region. However, the current built-in defenses are primarily sanity checks: NoGuard, random interventions, rule-based detection, oracle upper bound, and optional LLM or distilled baselines. Unless a submitted or simulator-trained defense produces robust positive $\Delta\alpha_c$ with acceptable cost, the leaderboard should not be the headline result. We therefore report the full defense leaderboard, per-scenario scores, threshold shifts, and cost-collapse tradeoffs in the appendix. This choice keeps the main paper focused on the scientific claim: harmful-agent scaling is measurable and structurally predictable.

Related Work

WolfBench is related to threshold models of collective behavior (????). Those models show how local adoption rules can produce global cascades. Our setting differs in two ways: harmful agents are an explicit controlled minority, and the measured object is a finite-size collapse-response curve in an interactive agent benchmark.

Financial network and contagion models study systemic risk through balance sheets, exposures, and network topology

(???). Market simulators such as ABIDES emphasize high-fidelity trading infrastructure (?). WolfBench is less realistic as a market simulator but more targeted as a safety instrument: it isolates how harmful population composition, social exposure, and defense policies move collapse thresholds.

Recent agent benchmarks evaluate tool use, web tasks, social interaction, and reward-safety tradeoffs (????). Harmful-agent benchmarks focus on whether individual agents complete unsafe tasks (?). WolfBench instead evaluates collective failures: harm emerges from coupled benign behavior after a harmful minority perturbs shared state.

Finally, our defense formulation is related to safe and constrained decision making (?), but the objective is different. A good WolfGuard policy is not simply a classifier with high recall. It must shift the critical harmful-agent ratio while preserving clean-market utility.

Discussion

The main implication is that agent safety should include society-level measurement. A system can appear safe under isolated-agent evaluation while remaining vulnerable to small coordinated minorities that alter shared state. In such settings, the relevant safety question is not only whether one harmful action is detected, but how close the whole society is to a critical regime. This is the dimension that single-agent harmfulness scores cannot capture: population composition can change the failure mode even when individual actions remain small.

Figure 4 placeholder: mechanism heterogeneity.

Panel A: $\alpha_c^m(N)$ for S1–S4 with confidence intervals.

Panel B: dominant channel annotations for each mechanism.

Panel C: transition-width or grid-coverage notes for S3/S4.

Panel D: compact table inset showing which mechanisms support clean logistic threshold estimation under the current grid.

Figure 4: Mechanism heterogeneity. WolfBench uses the same response-curve estimand across harmful mechanisms, but does not impose one global exponent or one universal threshold.

Figure 5 placeholder: structural threshold shifts.

Panel A: liquidity sweep, showing positive $\Delta\alpha_c$.

Panel B: graph degree / placement sweep, showing negative $\Delta\alpha_c$.

Panel C: retail susceptibility sweep, showing negative $\Delta\alpha_c$.

Panel D: feedback sweep, showing negative $\Delta\alpha_c$.

Panel E: sign-test summary with confidence intervals.

Figure 5: Structural threshold shifts. This figure tests the comparative-static direction predicted by the scaling model: parameters that dampen latent collapse pressure should shift the critical ratio right, while parameters that increase amplification should shift it left.

The finite-size framing also changes how defenses should be evaluated. A defense that slightly reduces average loss may be less meaningful than one that moves α_c to the right and widens the near-critical margin. Conversely, a defense that blocks many normal actions may reduce collapse while destroying utility. Threshold shift makes this tradeoff visible.

WolfBench should therefore be read as a measurement instrument. It is useful when its results are interpreted comparatively: across N , across harmful mechanisms, across structural parameters, and across defenses on the same critical-regime grid. It should not be interpreted as a market prediction engine.

Limitations and Ethical Considerations

WolfBench abstracts away many features of real markets: order-book detail, institutional participants, regulatory response, heterogeneous information, cross-asset portfolios, and strategic adaptation over long horizons. The 30-day horizon is a controlled stress window, not a claim that real harmful campaigns unfold on exactly that timescale. The finite-

size exponents reported in the paper are descriptive summaries of this benchmark, not universal market constants.

The benchmark also creates dual-use risk. We therefore describe harmful mechanisms at the level needed for scientific reproducibility, but avoid operational instructions that would improve real manipulation. Scenarios are case-inspired abstractions, not playbooks. Defense performance in WolfBench is evidence of benchmark behavior, not deployment readiness for real markets.

Finally, current evidence is strongest for the S1/S2 social-amplification mechanisms. S3 and S4 require broader or component-specific grids before making strong claims about universal finite-size trends. This is not a defect of the benchmark; it is precisely why mechanism-specific thresholds are reported.

Conclusion

We studied harmful-agent scaling: the way collapse probability changes as a harmful minority grows inside a finite agent society. The paper’s central claim is not that all

Lever	Theory sign for $\Delta\alpha_c$	Preferred measurement	Current diagnostic	Main-text status
Effective liquidity L	positive	threshold sweep over α	Exp8 liquidity $\Delta P(C)$	pending $\Delta\alpha_c$
Graph reach G	negative	degree sweep over α	Exp8 mean-degree sensitivity	pending $\Delta\alpha_c$
Harmful centrality	negative	placement sweep over α	Exp3/Exp8 placement effect	pending $\Delta\alpha_c$
Retail susceptibility β_s	negative	risk/social-beta sweep	Exp8 risk-appetite sensitivity	pending $\Delta\alpha_c$
Feedback ϕ	negative	feedback sweep near α_c	Exp4 fixed-alpha ablation	pending $\Delta\alpha_c$

Table 5: Comparative-statics sign-test plan. The table should be finalized with measured $\Delta\alpha_c$ and confidence intervals; current fixed-alpha diagnostics are weaker and should not be overstated.

financial systems exhibit a universal phase transition, nor that a particular simulator is the contribution by itself. The claim is that harmful-agent safety has a scaling dimension and that collective-risk evaluation needs estimands for finite societies: $\alpha_c(N)$, transition width, mechanism-specific thresholds, structural threshold shifts, and defense-induced threshold shift. WolfBench instantiates these estimands in a controlled financial agent society. This reframes agent safety from isolated-agent evaluation toward collective-risk measurement, where the central defense question is whether an intervention moves the society away from the critical regime without destroying normal behavior.

References